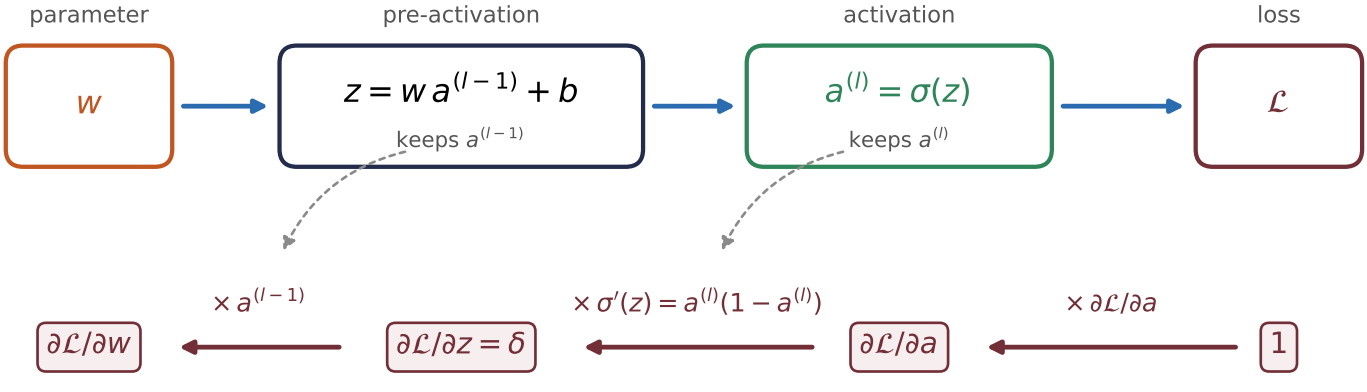
forward: compute each value and keep what the backward pass reuses
parameter
pre-activation
activation
loss
w
$z = w\,a^{(l-1)} + b$
keeps $a^{(l-1)}$
$a^{(l)} = \sigma(z)$
keeps $a^{(l)}$
$\mathcal{L}$
$\times\, a^{(l-1)}$
$\times\, \sigma'(z) = a^{(l)}(1 - a^{(l)})$
$\times\, \partial\mathcal{L}/\partial a$
$\partial\mathcal{L}/\partial w$
$\partial\mathcal{L}/\partial z = \delta$
$\partial\mathcal{L}/\partial a$
1
backward: multiply local derivatives, right to left